

GAUSS' LAW ON HYPERSURFACES

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ABSTRACT. Gauss' Law is an indispensable tool in Electrodynamics, allowing simple calculation of electric fields using Gaussian surfaces, such as a sphere, through surface integral formulae:

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enc}}.$$

In this paper, we generalise Gauss' Law to central forces of the form $\mathbf{F} = \frac{k\beta}{r^2} \hat{\mathbf{r}}$ where β is the physical quantity that induces the force, and discuss the high-dimensional equivalents.

1. GENERALISED GAUSS' LAW

We first begin with a discussion of Gauss' Law for central forces that obey the inverse-square law.

Theorem 1. *Suppose a particle with some physical quantity β situated at the origin induces a field (which obeys the principle of superposition) $\mathbf{K}$ of the form*

$$\mathbf{K} = \frac{k\beta}{r^2} \hat{\mathbf{r}}$$

where k is a constant independent of r .

Then, for any smooth closed surface S ,

$$\oint_S \mathbf{K} \cdot d\mathbf{a} = 4\pi k\beta_{\text{enc}}$$

Proof. Enclose the particle in a sphere S , of radius r . Then

$$\begin{aligned} \oint_S \mathbf{K} \cdot d\mathbf{a} &= \int_0^{2\pi} \int_0^\pi \left(\frac{k\beta}{r^2} \hat{\mathbf{r}} \right) \cdot ((r^2 \sin \theta d\theta d\phi) \hat{\mathbf{r}}) \\ &= k\beta \int_0^{2\pi} \int_0^\pi \sin \theta d\theta d\phi \\ &= 4\pi k\beta. \end{aligned}$$

The flux through the sphere is the same as the flux through an arbitrary C^1 surface S enclosing the particle by the Divergence Theorem, so that

$$\oint_{\mathcal{S}} \mathbf{K} \cdot d\mathbf{a} = 4\pi k\beta.$$

Note that particles outside $\mathcal{S}$ will not contribute to the overall field.

Now, consider an array of particles $\beta_1, \dots, \beta_n$. These particles will induce fields $\mathbf{K}_1, \dots, \mathbf{K}_n$. Our superposition assumption implies that the overall field is given by the sum $\mathbf{K} = \mathbf{K}_1 + \dots + \mathbf{K}_n$. Hence,

$$\begin{aligned} \oint_{\mathcal{S}} \mathbf{K} \cdot d\mathbf{a} &= \oint_{\mathcal{S}} \sum_{j=1}^n \mathbf{K}_j \cdot d\mathbf{a} \\ &= \sum_{j=1}^n \oint_{\mathcal{S}} \mathbf{K}_j \cdot d\mathbf{a} \\ &= \sum_{j=1}^n 4\pi k\beta_j \\ &= 4\pi k\beta_{\text{enc}}, \end{aligned}$$

which proves the theorem. $\square$

2. ON HYPERSURFACES

We shall extend Theorem 1 to a more general case, which is for central fields of the form $\mathbf{K} = \frac{k\beta}{r^n} \hat{\mathbf{r}}$, where n is a positive integer. Recall that the surface area element of an $(n-1)$ -dimensional sphere (in $\mathbb{R}^n$) of radius r is given by¹

$$dS^{n-1} = r^{n-1} \sin^{n-2}(\theta_1) \sin^{n-3}(\theta_2) \cdots \sin(\theta_{n-2}) d\theta_1 d\theta_2 \cdots d\theta_{n-1}.$$

We note that on the sphere, the normal vector is $\hat{\mathbf{n}} = \hat{\mathbf{r}}$. Therefore,

$$\begin{aligned} \int_{S^n(r)} \mathbf{K} \cdot d\mathbf{S} &= \int_0^{2\pi} \int_0^\pi \cdots \int_0^\pi \frac{k\beta}{r^n} r^n \sin^{n-1}(\theta_1) \sin^{n-2}(\theta_2) \cdots \sin(\theta_{n-1}) d\theta_1 d\theta_2 \cdots d\theta_n \\ &= k\beta \int_0^{2\pi} \int_0^\pi \cdots \int_0^\pi \sin^{n-1}(\theta_1) \sin^{n-2}(\theta_2) \cdots \sin(\theta_{n-1}) d\theta_1 d\theta_2 \cdots d\theta_n \\ &= k\beta \frac{2\pi^{\frac{n+1}{2}}}{\Gamma\left(\frac{n+1}{2}\right)}, \end{aligned}$$

where the last equality follows from the surface area of S^n ; see Proposition 2.54 in [Fol99].

For a smooth boundary $\mathcal{S}$ enclosing the origin, the flux through $\mathcal{S}$ will be the same as that through $S^n(r)$. Indeed, note that the field $\mathbf{K}$ has no divergence²[RT26]:

¹See page 293 of [SS03]

²Recall we are working in $\mathbb{R}^{n+1}$.

$$\nabla \cdot \mathbf{K} = \frac{1}{r^n} \frac{\partial}{\partial r} \left[r^n \frac{k\beta}{r^n} \right] = 0$$

Consider two piecewise smooth closed surfaces S_1 and S_2 enclosing the origin, and denote the regions enclosed by S_1 and S_2 by Ω_1 and Ω_2 , respectively. By the Divergence Theorem, we have³

$$\int_{\Omega_1} \nabla \cdot \mathbf{K} dV = \int_{S_1} \mathbf{K} \cdot d\mathbf{S}$$

and similarly for Ω_2 .

Defining the region between S_1 and S_2 by Ω , we have

$$\int_{S_2} \mathbf{K} \cdot d\mathbf{S} - \int_{S_1} \mathbf{K} \cdot d\mathbf{S} = \int_{\Omega} \nabla \cdot \mathbf{K} dV = 0.$$

In particular, we have

$$\int_{S_1} \mathbf{K} \cdot d\mathbf{S} = \int_{S_2} \mathbf{K} \cdot d\mathbf{S}.$$

In particular, we have, for a closed, smooth surface $\mathcal{S}$ enclosing the origin:

$$\begin{aligned} \int_{\mathcal{S}} \mathbf{K} \cdot d\mathbf{S} &= \int_{S^n(r)} \mathbf{K} \cdot d\mathbf{S} \\ &= k\beta \frac{2\pi^{\frac{n+1}{2}}}{\Gamma\left(\frac{n+1}{2}\right)}. \end{aligned}$$

We are now ready to prove the main result in this paper.

Theorem 2. *Suppose a particle with some physical quantity β situated at the origin induces a field (which obeys the principle of superposition) $\mathbf{K}$ of the form*

$$\mathbf{K} = \frac{k\beta}{r^n} \hat{\mathbf{r}}$$

where k is a constant independent of r , and n is a positive integer.

Then, for any C^1 , closed smooth surface $\mathcal{S}$,

$$\int_{\mathcal{S}} \mathbf{K} \cdot d\mathbf{S} = k\beta_{enc} \frac{2\pi^{\frac{n+1}{2}}}{\Gamma\left(\frac{n+1}{2}\right)}.$$

³See page 17 of [Hun14].

Proof. Consider an array of particles with $\beta_1, \dots, \beta_n$ enclosed by a C^1 surface $\mathcal{S}$. Then, the overall field is given by

$$\mathbf{K} = \sum_{j=1}^n \mathbf{K}_j,$$

which follows from our superposition assumption. Then,

$$\begin{aligned} \int_{\mathcal{S}} \mathbf{K} \cdot d\mathbf{S} &= \int_{\mathcal{S}} \sum_{j=1}^n \mathbf{K}_j \cdot d\mathbf{S} \\ &= \sum_{j=1}^n \int_{\mathcal{S}} \mathbf{K}_j \cdot d\mathbf{S} \\ &= \sum_{j=1}^n k\beta_j \frac{2\pi^{\frac{n+1}{2}}}{\Gamma\left(\frac{n+1}{2}\right)} \\ &= k\beta_{\text{enc}} \frac{2\pi^{\frac{n+1}{2}}}{\Gamma\left(\frac{n+1}{2}\right)}, \end{aligned}$$

which is the result we wanted to prove. $\square$

REFERENCES

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